

TA212 PROJECT 2024-25(I)

GRASSCUTTER

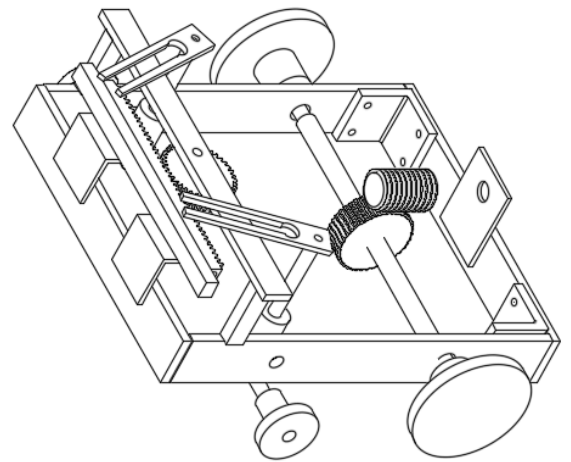
Group - G9

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INTRODUCTION :

Objective : The Grass Cutter Project aims to design and develop an efficient and automated system for maintaining lawns and green spaces. This project focuses on creating a solution that minimizes human effort while maximizing performance with low power consumption and also ensures that the design is simple and comes with a low cost of construction.

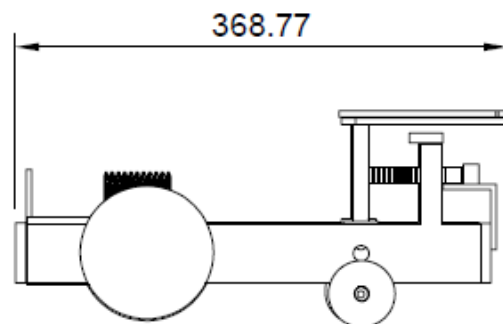
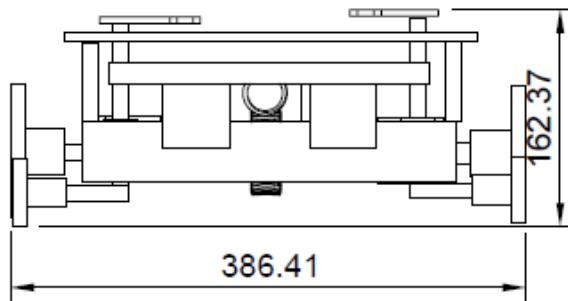
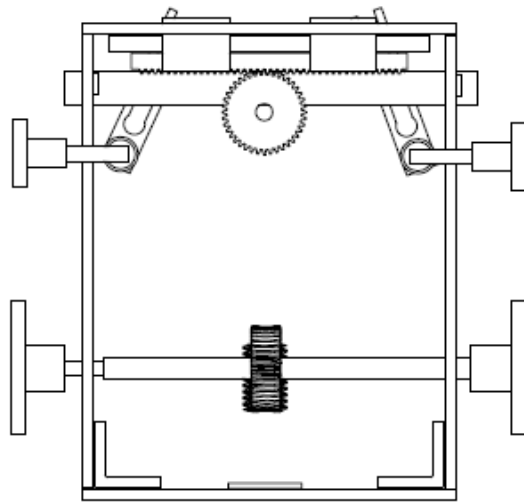
How many parts were manufactured : 16

Cost of the project : Rs 13175

Improvements that could be made :

- 1) High speed motors can be used instead of high torque.
- 2) Sensors could be installed in order to detect grass.
- 3) We can also make it move in all the 4 directions using a controller.
- 4) 2-3 more blades can be used so that it can cut in many directions at one time.

ASSEMBLY DRAWING



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LABELED DIAGRAM OF THE ASSEMBLY

Parts List			
Item	Qty	Part Number	Material
1	1	Frameback	Generic
2	1	Framefront	Generic
3	2	Frameside	Generic
4	2	Clamp 1	Generic
5	2	Part5_front_rod	Generic
6	1	Rack 212 2	Generic
7	1	Pinion_final[1]	Steel
8	2	Front wheel	Generic
9	2	Part2_front_collor	Generic
10	1	Backwheel1	Generic
11	1	Backwheel2	Generic
12	1	Shaft	Generic
13	1	Motor support	Generic
14	2	Part1_front_arm	Generic
15	1	Worm Gear	Steel, Mild
16	1	Worm	Steel, Mild
17	1	Wormgearsupport	Generic

The diagram shows an exploded view of a mechanical assembly. The components are numbered 1 through 17, corresponding to the parts list. The assembly includes a frame (1, 2, 3), clamps (4), rods (5), a rack and pinion system (6, 7), front wheels (8), front collars (9), back wheels (10, 11), a shaft (12), a motor support (13), front arms (14), a worm gear (15), a worm (16), and a worm gear support (17). The diagram illustrates the spatial relationship and assembly sequence of these parts.

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Torque Calculations

Given Data

- Vehicle mass, $m_1 = 12$ kg
- Wheel outer diameter, $D_1 = 100$ mm
- Wheel mass, $m_{D1} = 0.4$ kg per piece
- Number of wheels, $n_1 = 2$
- Rolling friction coefficient, $\mu_1 = 0.035$
- Drive mechanism efficiency, $\eta = 0.85\%$
- Floor slope, $\alpha = 0^\circ$
- Operating speed, $V_1 = 0.2$ m/min
- Safety factor, $S.F = 1.5$
- Modulus of worm gear, $m = 1.5$
- Number of teeth in the worm gear, $Z_w = 40$
- Number of threads in the worm, $Z_t = 1$
- Gear ratio=1:40

Calculated Results

1. Required Speed

$$\begin{aligned} V_m &= \frac{V_1}{\pi D_1 \times 10^{-3}} \times \text{gear ratio} \\ &= \frac{0.2}{3.14 \times 100 \times 10^{-3}} \times \frac{1}{40} = 25.48 \text{ r/min} \end{aligned}$$

2. Required Torque

$$\begin{aligned} F &= 9.8 ((m_1 + n_1 \times m_{D1}) \times (\sin \alpha + \mu_1 \cos \alpha)) \\ &= 9.8 ((12 + 2 \times 0.4) \times (\sin 0 + 0.035 \cos 0)) = 4.390 \text{ N} \end{aligned}$$

$$T_L = \frac{F \times D_1 \times 10^{-3}}{2\eta \times 0.01} \times \frac{D_{p1}}{D_{p2}}$$

$$= \frac{4.390 \times 100 \times 10^{-3}}{2 \times 0.85 \times 0.01} \times \frac{1}{40} = 0.6456 \text{ N} \cdot \text{m}$$

Total Required Torque T :

$$T = T_L \times \text{Safety Factor}$$

$$= 0.6456 \times 1.5 = 0.9684 \text{ N} \cdot \text{m}$$

Force Calculations of Pinion Data:

- Normal module, $m = 1.5\text{mm}$
- Number of teeth on pinion, $Z_1 = 40$
- Outside diameter of pinion, $D_{\text{pinion}} = 63\text{mm}$

Mass Calculation Data:

- Rack dimensions: $207 \times 12 \times 12 \text{ mm}$
- Rack Holder dimensions: $216 \times 15 \times 15 \text{ mm}$
- Material: Mild steel (Density $\rho = 7.85 \text{ g/cm}^3$)

Mass Calculation for Rack and Plate:

The mass of the rack and square plate is calculated using their volumes and the density of mild steel.

$$\text{Volume of rack} = 12 \times 12 \times 207 = 29664 \text{ mm}^3 = 29.664 \text{ cm}^3$$

$$\text{Mass of rack} = 7.85 \times 29.664 = 232.3 \text{ g}$$

The mass of the rack holder:

$$\text{Volume of plate} = 216 \times 15 \times 15 = 48600.5 \text{ mm}^3 = 48.6 \text{ cm}^3$$

$$\text{Mass of plate} = 7.85 \times 48.6 = 381.5 \text{ g}$$

The total mass is the sum of the mass of the rack, the rack holder:

$$\text{Total mass} = 232.3 + 381.5 \text{ g} = 613.8 \text{ g}$$

Force Calculation

The force due to gravity acting on the mass is given by:

$$F = m \times g$$

Where m is the total mass in kilograms and $g = 9.81 \text{ m/s}^2$.

First, convert the total mass to kilograms:

$$m = 613.8 \text{ g} = 0.6138 \text{ kg}$$

$$\text{Thus Force acting} = 0.6138 \times 9.81 = 6.021 \text{ N.}$$

COST ANALYSIS

Cost of material:

1) Weight of Mild Steel = 12.9 kg

Cost = $12.9\text{kg} \times 100\text{Rs/kg} = \text{Rs. } 1,290$

3) Cost of Motor = $\text{Rs. } 600 \times 2 = \text{Rs. } 1,200$

4) Cost of blade = $\text{Rs. } 100$

5) Cost of motor holder = $\text{Rs. } 50 \times 2 = \text{Rs. } 100$

6) Cost of electric kit = $\text{Rs } 1000$

Cost of processes

1) **Milling:** 1 Worm and 1 Rack and Pinion gear : $2 + 2 \text{ hrs} = 4\text{hrs} \rightarrow \text{cost} = 4\text{hrs} \times 250\text{Rs./hr} = \text{Rs. } 1,000$

2) **Turning:** $4 \text{ hrs} \rightarrow \text{cost} = 4 \text{ hrs} \times 150\text{Rs./hr} = \text{Rs. } 600$

3) **Drilling:** $5 \text{ hrs} \rightarrow \text{cost} = 5 \text{ hrs} \times 75\text{Rs./hr} = \text{Rs. } 375$

Labor costs

5 Labs $\rightarrow 2.5 \text{ hr in each lab} \rightarrow 12.5 \text{ hrs} \rightarrow \text{Rs } 600 \text{ for } 1 \text{ hr} \rightarrow \text{Rs } 7500$

Total cost = Rs. 13165

